



DentMAG: Masked Autoencoding with Guidance Cues for Comprehensive Dental Diagnosis



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Introduction

Current AI in dentistry apply separate models for isolated tasks, lacking holistic clinical reasoning. We introduce DentMAG, a unified model that detects multiple oral conditions simultaneously including caries, gingivitis and dental plaque from intraoral photos. This unified pipeline mirrors the clinical diagnostic workflow by providing a holistic analysis that transitions from oral hygiene screening (plaque levels) through gingivitis assessment, to pixel-level caries segmentation, enabling a comprehensive understanding of oral health in a single forward pass.

Objectives

- To establish a large-scale, smartphone-captured intraoral image dataset specifically targeting adolescents;
- To design a multi-task and multi-scale AI model optimized for fine-grained feature extraction of subtle oral characteristics.

Study Design

- **Study Cohort:** 1000+ local secondary school students (aged 12-18) from 8 secondary schools across three areas of Hong Kong.
- **Intraoral Photo Acquisition:** Intraoral imaging via mainstream smartphones, utilizing on-screen gridline guidance and mobile torch to capture five standard views: frontal, left/right buccal and upper/lower occlusal
- **Automated Preprocessing:** Intelligent region-of-interest (ROI) extraction powered by vision foundation model to ensure data consistency.
- **Standardized Annotation:** Clear definition of computer vision tasks, including specific tooth inclusion/exclusion criteria and standardized diagnostic criteria

Standard Image Acquisition

- Mainstream smartphones (IOS, Android etc.,)
- On-screen grid-line assistance for tooth centering and focal precision.
- External dental headlamps or mobile torch to ensure consistent lighting.
- Hybrid approach using retractors (for limited opening) and non-retractor captures.

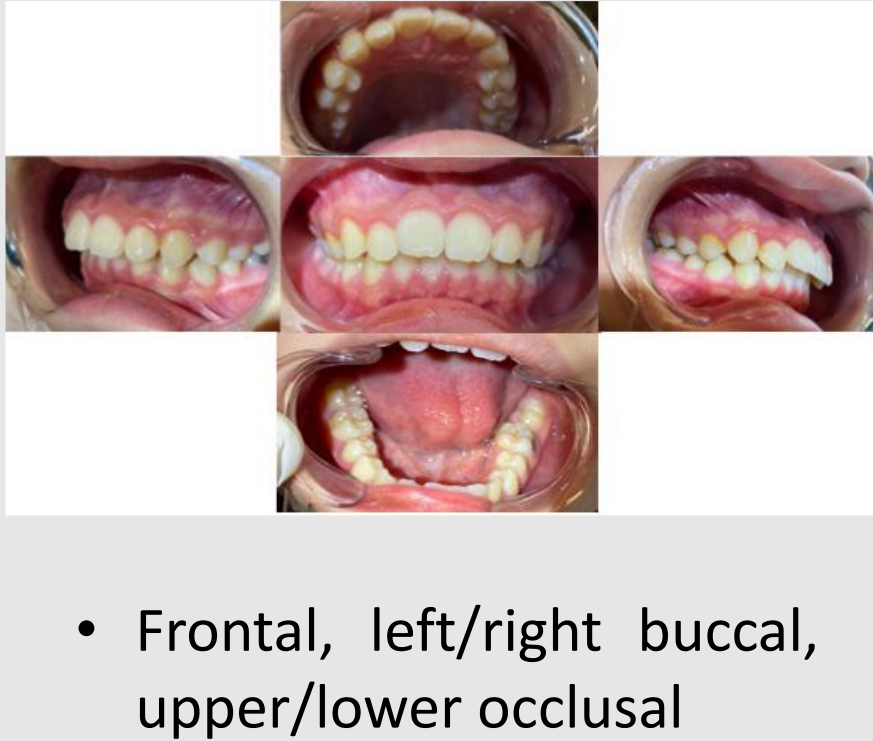
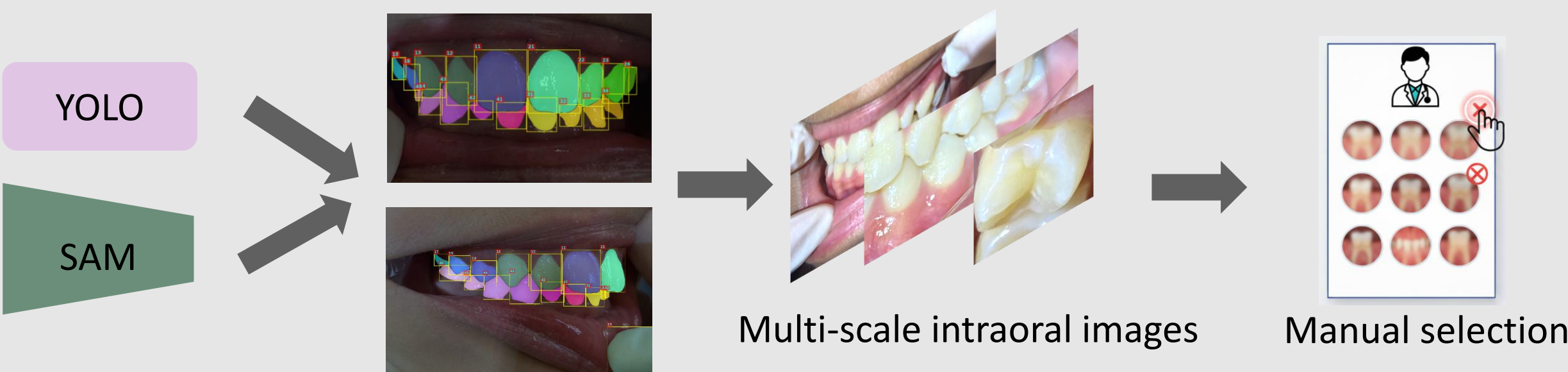


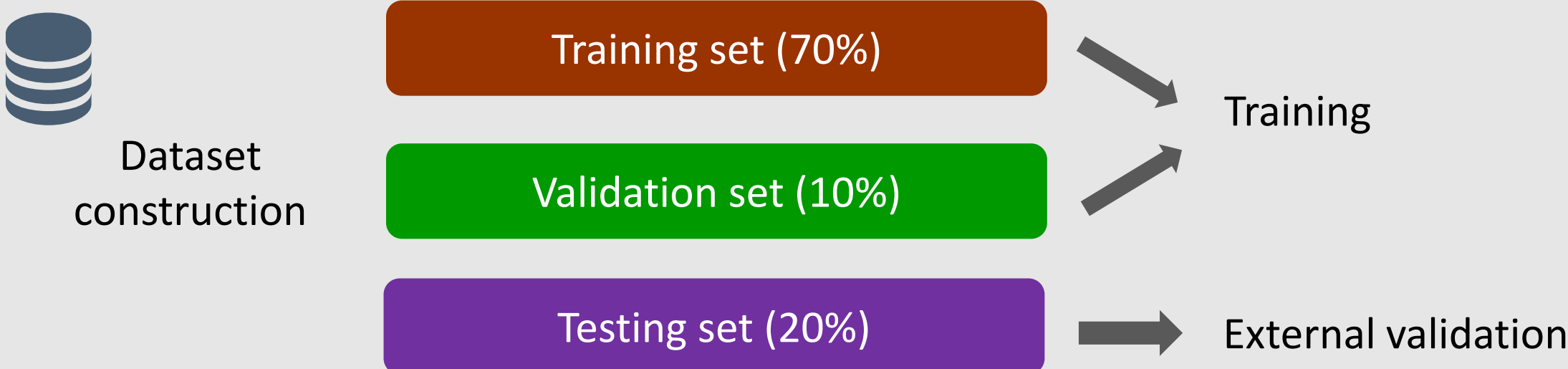
Image Preprocessing

- Segment Anything (SAM) for oral cavity extraction
- YOLO for single tooth detection

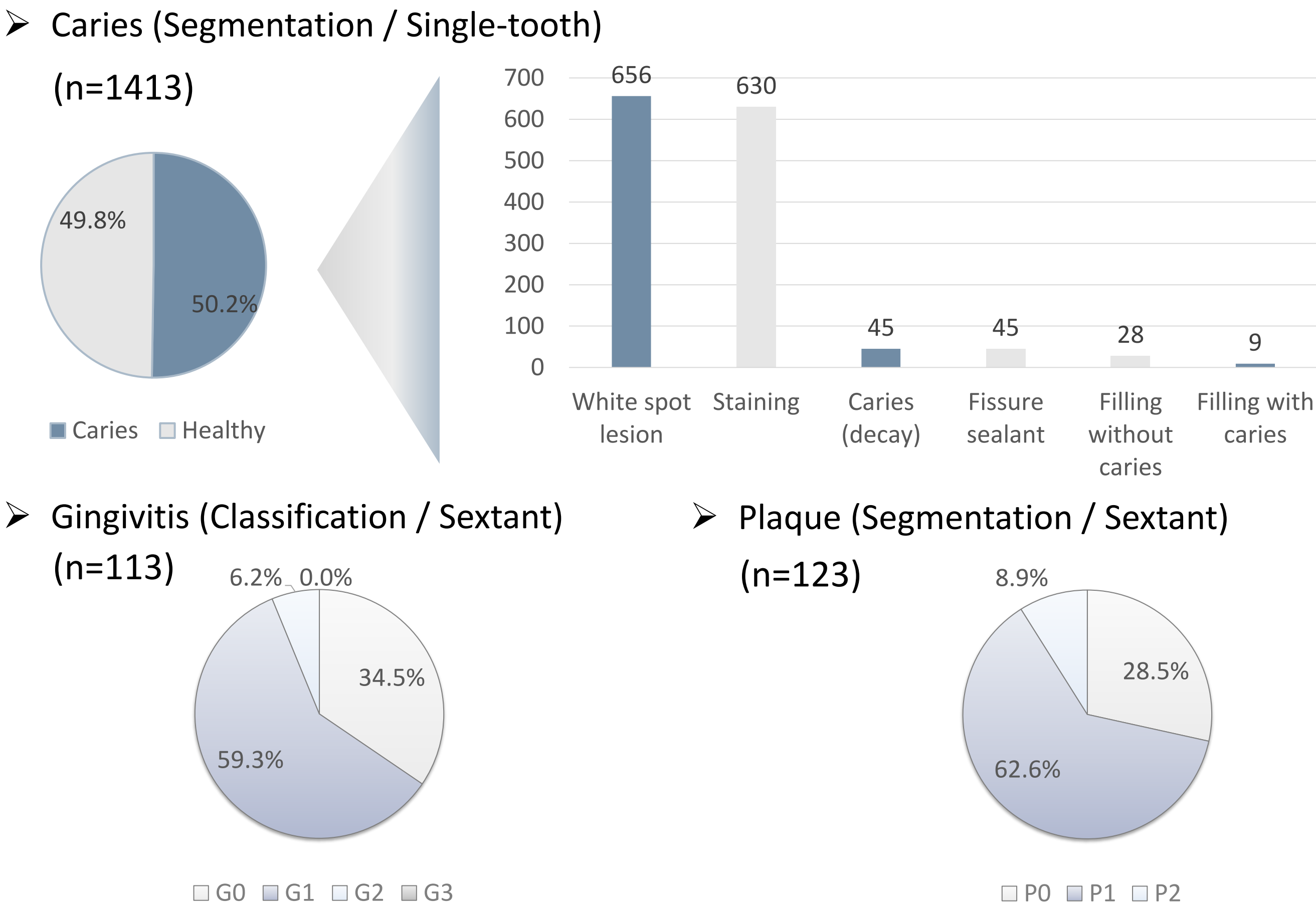


Annotation

	Caries	Gingivitis	Plaque
Computer vision task	Segmentation	Classification	Segmentation
Level of image scale	Single tooth	Sextant	Sextant
Capture view	Frontal, left/right, upper/lower	Frontal, left/right	Frontal, left/right
Reference standard	ICDAS	MGI (Modified Gingival Index)	VPI (Visible Plaque Index)



Annotated Data Statistics



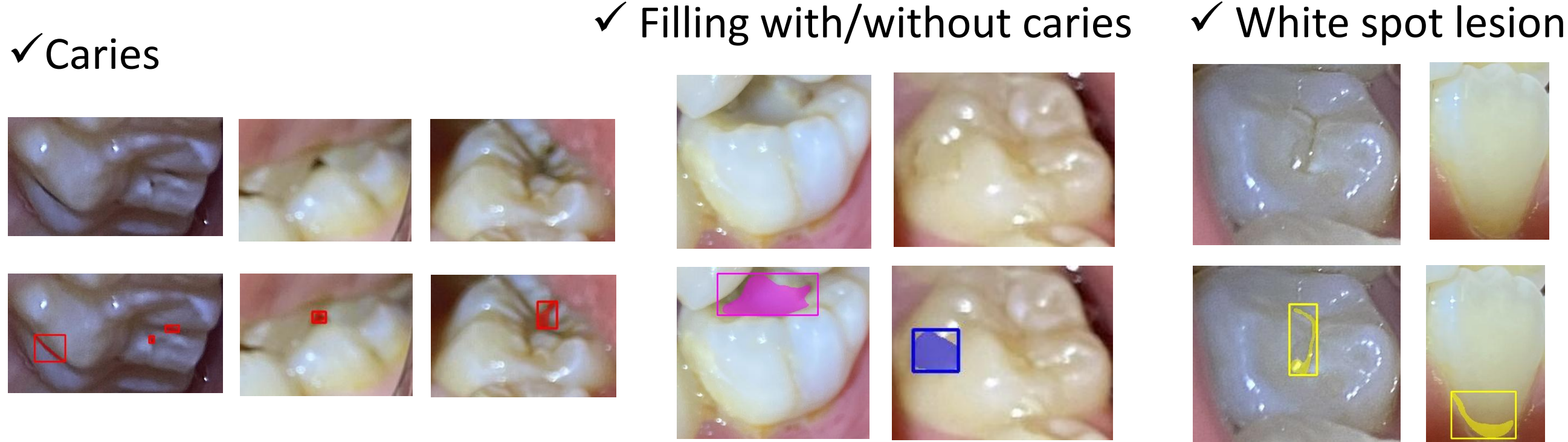
- **Caries Early Detection:** Considerable cases of white spot lesion underscores the critical window for early intervention of adolescents especially on young permanent teeth.
- **Gingival Health Status:** Mild to severe gingiva affected (G1+G2+G3=65.5%), indicating gingivitis are prevalent among this age group.
- **Plaque Accumulation:** Only 28.5% of the studied sextants were found to be free of plaque on tooth surfaces, indicating that toothbrushing among adolescents is often ineffective.

Preliminary Results and Discussion

➤ Caries (n=1413)

Task	AP	Pathological Category	Detection AP (%)	Segmentation AP (%)	Sensitivity	F1-score
Object Detection	90.3	Filling (No caries)	96.5	90.1	0.96	0.94
		Filling (With caries)	92.8	91.0	0.94	0.92
Instance Segmentation	80.0	Fissure sealant	91.6	70.3	0.90	0.89
		Caries	87.2	67.8	0.91	0.86
		White spot lesion	77.9	58.6	0.91	0.80

Visualization: GT vs. prediction



- The baseline model successfully captures core pathological features without requiring massive optimization.
- Highly accurate at identifying approximate locations of both cavitated and non-cavitated caries in nonclinical settings.
- Detection significantly outperforms Segmentation. The model finds them easily but struggles with pixel-perfect boundaries due to fuzzy lesion edges.

➤ Gingivitis (n=113)



Conclusion

DentMAG provides a unified framework that streamlines fragmented dental AI tasks into a cohesive multi-task computer vision system. By standardizing the entire pipeline, from data acquisition and preprocessing to multi-scale annotation and training, DentMAG has the potential for oral hygiene and oral health screening from users uploaded intraoral photos.